Two-Column Research Paper

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Abstract

This paper presents a two-column layout commonly used in academic publications. The converter should correctly identify column boundaries and reconstruct the reading order so that text flows from the left column to the right column on each page, not interleaving lines from both columns.

1 Introduction

Academic papers frequently use two-column layouts to maximize information density on each page. This presents a challenge for PDF extraction because the converter must determine which text blocks belong to which column.

The spatial analysis must consider the horizontal gap between columns as a separator. Text blocks on the left side of the gap should be read first, followed by text blocks on the right side.

2 Methodology

Our approach uses a vertical split-point detector. We analyze the distribution of text block x-coordinates and identify a gap region near the horizontal center of the page. Blocks to the left of this gap are assigned to column 1, and blocks to the right are assigned to column 2.

Within each column, blocks are ordered top-to-bottom. The final reading order concatenates column 1 followed by column 2 for each page.

3 Results

Testing on 500 academic papers showed that our column detection achieves 97.3% accuracy in correctly ordering text blocks. The most common failure case involves wide equations or figures that span both columns.

4 Conclusion

Two-column layout handling is essential for academic paper conversion. Our spatial analysis approach provides robust column detection without requiring machine learning or training data.